

Revision date 19-Nov-2024

Revision Number 2

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** 4X Laemmli Sample Buffer

**Catalogue Number(s)** 1610747

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
u1A, 62 Ferndell Street,  
South Granville NSW 2142  
Australia

For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

**Serious eye damage/eye irritation**

Category 1 - (H318)

### Label elements

Corrosion



**Signal word**

Danger

**Hazard statements**

H318 - Causes serious eye damage

**Precautionary Statements - Prevention**

Wear protective gloves/protective clothing/eye protection/face protection

**Precautionary Statements - Response**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTRE or doctor

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

No information available.

**SECTION 3: Composition/information on ingredients****Substance**

Not applicable

**Mixture**

| Chemical name                                  | CAS No.     | Weight-%    |
|------------------------------------------------|-------------|-------------|
| 1,2,3-Propanetriol                             | 56-81-5     | 35 - 50     |
| Sulfuric acid, monododecyl ester, lithium salt | 2044-56-6   | 2.5 - 5     |
| Hydrochloric acid                              | 7647-01-0   | 0.1 - 0.249 |
| Non-hazardous ingredients                      | Proprietary | Balance     |

**SECTION 4: First aid measures****Description of first aid measures****General advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Emergency telephone number**Poisons Information Centre, Australia: 13 11 26  
Poisons Information Centre, New Zealand: 0800 764 766**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Eye contact**

Get immediate medical advice/attention. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area.

**Skin contact**

Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Call a doctor.

**Self-protection of the first aider**

Avoid contact with skin, eyes or clothing. Wear personal protective clothing (see section 8).

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Burning sensation. Prolonged contact may cause redness and irritation.

**Indication of any immediate medical attention and special treatment needed**

**Note to doctors** Treat symptomatically.

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** None known.

**Special protective actions for fire-fighters**

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

**SECTION 6: Accidental release measures****Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Use personal protective equipment as required.

**Other information** Refer to protective measures listed in Sections 7 and 8.

**For emergency responders** Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions** Prevent further leakage or spillage if safe to do so.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**SECTION 7: Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product.

**General hygiene considerations** Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children.

**Incompatible materials** Strong acids. Strong bases. Strong oxidising agents.

## **SECTION 8: Exposure controls/personal protection**

**Control parameters**

**Exposure Limits**

| Chemical name                  | Australia                                  | ACGIH TLV      |
|--------------------------------|--------------------------------------------|----------------|
| 1,2,3-Propanetriol<br>56-81-5  | TWA: 10 mg/m <sup>3</sup>                  |                |
| Hydrochloric acid<br>7647-01-0 | Peak: 5 ppm<br>Peak: 7.5 mg/m <sup>3</sup> | Ceiling: 2 ppm |

**Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Tight sealing safety goggles.

**Skin and body protection** Wear suitable protective clothing.

**Hand protection** Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## **SECTION 9: Physical and chemical properties**

**Information on basic physical and chemical properties**

**Physical state** Liquid  
**Appearance** aqueous solution  
**Colour** blue  
**Odour** Odourless.  
**Odour threshold** No information available

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 6.8               |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | > 100 °C          |                         |
| Flash point                             | > 160 °C          |                         |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapour pressure                         | No data available | None known              |
| Relative vapour density                 | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | 392.8 °C          |                         |
| Decomposition temperature               |                   | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |
| Explosive properties                    | Not applicable    |                         |
| Oxidising properties                    | Not applicable    |                         |
| <u>Other information</u>                |                   |                         |
| Molecular weight                        | Not applicable    |                         |
| VOC content                             | Not applicable    |                         |

## SECTION 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials Strong acids. Strong bases. Strong oxidising agents.

### Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

## SECTION 11: Toxicological information

### Acute toxicity

### Information on likely routes of exposure

#### Product Information

|                     |                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available.                                                                                  |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye damage. May cause irreversible damage to eyes.                |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Causes mild skin irritation.                                                     |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhoea. |
| <b>Symptoms</b>     | Redness. Burning. May cause blindness. Prolonged contact may cause redness and irritation.                                                         |

### Numerical measures of toxicity - Product Information

No information available

The following values are calculated based on chapter 3.1 of the GHS document

|                        |                  |
|------------------------|------------------|
| <b>ATEmix (oral)</b>   | 22,740.30 mg/kg  |
| <b>ATEmix (dermal)</b> | 174,440.90 mg/kg |

#### Component Information

| Chemical name      | Oral LD50               | Dermal LD50             | Inhalation LC50         |
|--------------------|-------------------------|-------------------------|-------------------------|
| 1,2,3-Propanetriol | = 12600 mg/kg ( Rat )   | > 10 g/kg ( Rabbit )    | > 2.75 mg/L ( Rat ) 4 h |
| Hydrochloric acid  | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit ) | = 1.68 mg/L ( Rat ) 1 h |

See section 16 for terms and abbreviations

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                                                                    |
|------------------------------------------|------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | May cause skin irritation. Classification based on data available for ingredients. |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes burns.              |
| <b>Respiratory or skin sensitisation</b> | Based on available data, the classification criteria are not met.                  |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met.                  |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met.                  |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met.                  |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met.                  |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met.                  |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met.                  |

**SECTION 12: Ecological information****Ecotoxicity****Ecotoxicity** Harmful to aquatic life.

| Chemical name      | Algae/aquatic plants | Fish                                         | Toxicity to microorganisms | Crustacea |
|--------------------|----------------------|----------------------------------------------|----------------------------|-----------|
| 1,2,3-Propanetriol | -                    | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss) | -                          | -         |

**Persistence and degradability****Persistence and degradability** No information available.**Bioaccumulative potential****Bioaccumulation** There is no data for this product.**Component Information**

| Chemical name      | Partition coefficient |
|--------------------|-----------------------|
| 1,2,3-Propanetriol | -1.75                 |

**Mobility****Mobility in soil** No information available.**Mobility** No information available.**Other adverse effects****Other adverse effects** No information available.**SECTION 13: Disposal considerations****Disposal methods****Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.**Contaminated packaging** Do not reuse empty containers.**SECTION 14: Transport information****ADG** Not regulated**IATA** Not regulated**IMDG** Not regulated**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

## SECTION 15: Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

##### Australia

See section 8 for national exposure control parameters

##### **Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

Classified as a scheduled poison according to the Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)

**Poison Schedule Number** 6

##### **Major hazard (accident/incident planning) regulation**

Verify that licence requirements are met

##### Named hazardous chemicals

| Chemical name                 | Threshold quantity (T)                                     |
|-------------------------------|------------------------------------------------------------|
| Hydrochloric acid - 7647-01-0 | 250 tonne TQ anhydrous<br>250 tonne TQ refrigerated liquid |

##### **National pollutant inventory**

Subject to reporting requirement

| Chemical name                 | National pollutant inventory                                                                                                                                                                                       |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrochloric acid - 7647-01-0 | 10 tonne/yr Threshold category 1<br>400 tonne/yr Threshold category 2a<br>1 tonne/h Threshold category 2a<br>2000 tonne/yr Threshold category 2b<br>60000 MWH Threshold category 2b<br>20 MW Threshold category 2b |

##### **International Inventories**

Contact supplier for inventory compliance status

##### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

## SECTION 16: Other information

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 19-Nov-2024

**Revision Note** Reformatted and updated existing information.

##### **Key or legend to abbreviations and acronyms used in the safety data sheet**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| C       | Carcinogen                  |      |                                  |



**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

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**End of Safety Data Sheet**